



Correlation of properties with radiation response in emerging scintillator materials such as Hydrogels, Nanocomposites and CaF_2

Team Scintilaleins

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1. Our Motivation & Purpose of Visiting CERN

We are passionate about applying material sciences to address global challenges. Through our research, we aim to accelerate the development of radiation therapy and medical imaging tools. We believe our team's diversity in subject and nationality, combined with CERN's expertise, can help us turn these novel materials, like hydrogels, into a standard for healthcare technology.

During the preliminary literature review, it became evident that while plastic scintillators have been the industry standard, alternative scintillating materials like nanocomposites and hydrogels, present applications, that have not been explored. To address this gap, our study explores effectiveness of novel scintillating materials.

Beyond our immediate research goals, we see the chance to visit CERN as an opportunity to expand our scientific knowledge. We hope to develop a deeper understanding of physics and its applications and cultivate a professional network that will support our future contributions.

2. Introduction

High-energy radiation detection relies on materials that convert deposited energy into detectable ultraviolet or visible photons via scintillation. Traditional detectors use plastic scintillators or inorganic crystals for light yield but possess limitations. These challenges have increased interest in alternative materials, such as nanocomposite-based scintillators, which are promising yet remain underexplored in high energy physics, leaving a gap in literature.

This study addresses that gap while also introducing a further alternative: Hydrogel-based scintillators. Hydrogels are characterised by their optical transparency and their chemical compositions. They can also easily mimic the human tissue, which offers significant potential for radiation dosimetry.

A key aspect that also remains unexplored is how thickness can impact scintillation. A further aim of this study is to investigate how scintillator performance can vary with thickness.

2.1 Aim

This study investigates the scintillation performance of Agarose-based Hydrogels, Nanocomposites, CaF₂ crystals, and plastic scintillators across different thicknesses by comparing their light output and signal response to evaluate potential low-cost radiation detection materials.

3. Theory

Scintillating detectors convert the deposited energy by incident gamma radiation into visible photons, which are then collected by a photomultiplier tube (PMT) and converted into measurable electrical signals. To calculate the light yield, the Relative Light and the Photoelectron Yield (pe/ MeV) are calculated.

The Relative Light Yield is given by the formula:

$$Relative\ Light\ Yield = \frac{Centroid_{sample}}{Centroid_{Reference}} \times 100$$

Photoelectron Yield is given by the formula:

$$Y_{pc} = \left(\frac{2.355}{R}\right)^2 \times \frac{1}{E}$$

Where R is calculated with the equation:

$$R = \frac{FWHM}{E} \times 100$$

Where:

R : Resolution

$FWHM$: Full width at half maximum

E : Energy of the channel (MeV)

4. Methodology

4.1 Preliminary Experiment

The preliminary experiment focuses on the preparation of Hydrogels and Nanocomposites. The agarose-based hydrogels were prepared following a standard hydrogel preparation method. To enhance the light yield, Fluorescein Sodium was added.

The ZnO Nanocomposites were synthesized using a standard procedure. All samples were cut to $3\text{cm} \times 3\text{cm}$ with thicknesses ranging from 1 to 10mm.



Figure 4.1.1 Hydrogel Sample



Figure 4.1.2 Nanocomposite Samples while curing

4.2 Main Experiment

For the primary experiment, all samples were placed on an acrylic window. To ensure maximum light collection and prevent external interference, each sample was covered in aluminium foil and black PVC tape with one side uncovered. Optical grease was applied to the uncovered side prior to coupling to the PMT. From the PMT, spectral data were acquired using the Genie 2000 Gamma Acquisition and Analysis Software, from which both the light yield and photoelectron yield were calculated.



Figure 4.2.1 Experimental Setup

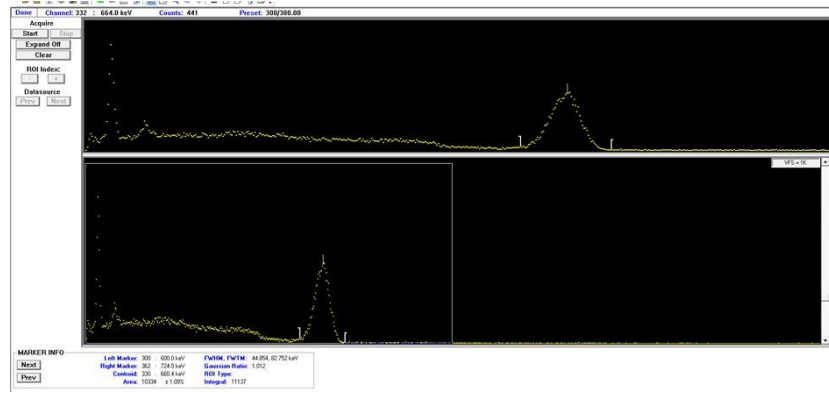


Figure 4.2.2 Spectrum from the Genie 2000 Gamma Acquisition and Analysis Software

4.3 Simulation Method

Simulation Python Code

A Monte-Carlo simulation was programmed to predict the interactions of 2 GeV muons with four different scintillator samples. The thicknesses of the modelled slabs ranged from 1 to 10 mm with the purpose of quantifying the deposition of energy, quenched photon production, and relative light yield. In this simulation, the energy loss of muons was calculated using the Bethe-Bloch equation:

$$\langle -\frac{dE}{dx} \rangle = Kz^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 W_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right]$$

Where,

$\frac{dE}{dx}$: Mean energy loss per unit path length

K : Constant = $4\pi N_A r_e^2 m_e c^2 = 0.307 \text{ MeV cm}^2/\text{g}$

z : Charge of the incoming particle (for muons, $z = 1$)

Z : Atomic number of the absorber material

A : Atomic mass of absorber

β : Particle velocity / speed of light

γ : Lorentz factor = $1/\sqrt{1 - \beta^2}$

m_e : Electron mass

c : Speed of light

$W_{\max}$: Maximum energy transfer in a single collision

I : Mean excitation potential of the material

$\delta(\beta\gamma)$: Density effect correction

with Sternheimer density-effect corrections of the form:

$$\delta = \begin{cases} 4.606x + C + a(x_1 - x)^m, & x_0 < x < x_1 \\ 4.606x + C, & x > x_1 \\ 0, & \text{otherwise} \end{cases}.$$

Where:

x : $\log_{10}(\beta\gamma)$

C : Material Constant

a, m : Fitting parameters for density correction

x_0, x_1 : Material-dependent parameters

Additionally, the stochastic ionization fluctuations were estimated using Landau straggling's Gaussian representation. Lastly, photon production was estimated using Birks' law to account for quenching effects:

$$\frac{dL}{dx} = \frac{S \frac{dE}{dx}}{1 + kB \frac{dE}{dx}}$$

Where:

$\frac{dL}{dx}$: Scintillation light produced per unit length

S : Scintillation efficiency

kB : Birk's constant (quenching factor)

$\frac{dE}{dx}$: Energy loss from Bethe-Bloch

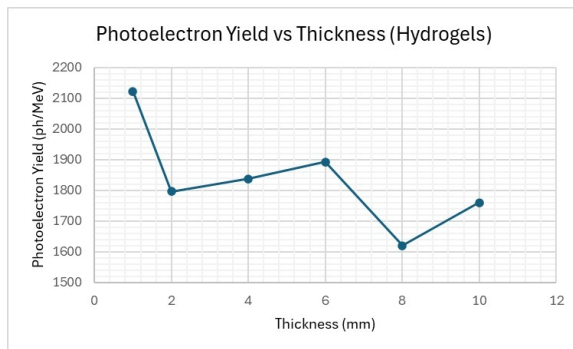
4.4 Proposed Method for CERN

As an extension of this study, we propose utilising a muon beam energy of approximately 2GeV at CERN. Muons are Minimum Ionizing Particles (MIPs), making them suitable for comparing the scintillation response of different materials.

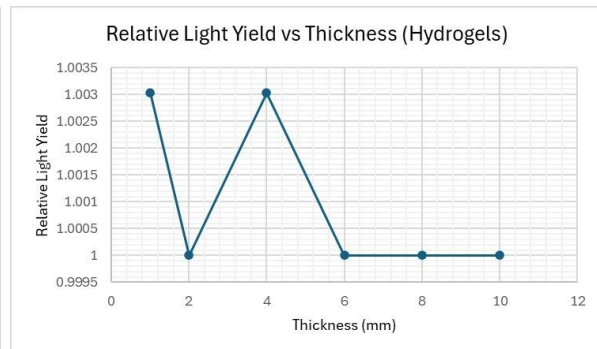
The prepared samples will be placed in the beam path together with the reference detector. Scintillation photons that are generated during the muon transit will be collected by the PMTs and processed through an ADC. By integrating the results, the data can be analysed.

5. Analysis

5.1 Experimental Results



*Figure 5.1.1: Photoelectron Yield
vs
Thickness for hydrogels*



*Figure 5.1.2: Relative Light Yield
vs
Thickness for hydrogels*

Hydrogels: The maximum photoelectron yield of 2100ph/MeV is achieved at 1mm, followed by a gradual attenuation as seen in *Figure 5.1.1*. This trend is corroborated by the relative light yield, which peaks at 1mm before stabilizing, as shown in *Figure 5.1.2*. Since the optimal thickness was deduced to 1mm, it can be suitable for radiation protection.

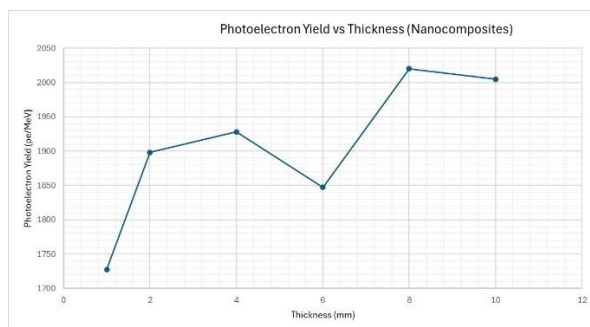


Figure 5.1.3: Photoelectron Yield
vs
Thickness for nanocomposites

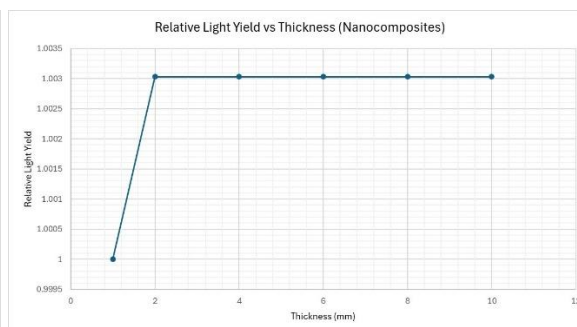


Figure 5.1.4: Relative Light Yield
vs
Thickness for nanocomposites

Nanocomposites: The relative light yield demonstrates a sharp increase when moving from 1mm to 2mm, after which the yield remains almost perfectly plateaued at ~ 1.003 , as shown in Figure 5.1.4. This suggests that the light yield efficiency reaches a saturation beyond 2 mm thickness. Conversely, the absolute photoelectron yield shows a positive correlation, as shown in Figure 5.1.3, which could be associated with the experimental variation.

5.2 Simulation Results

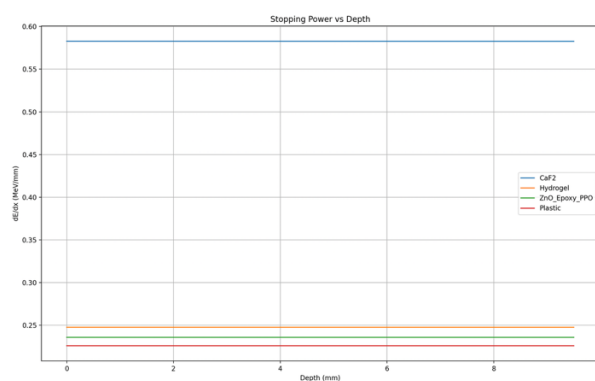


Figure 5.2.1 Stopping Power vs Depth

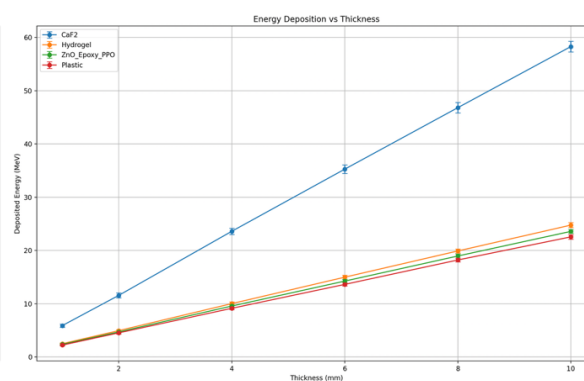


Figure 5.2.2 Energy Deposition vs Thickness

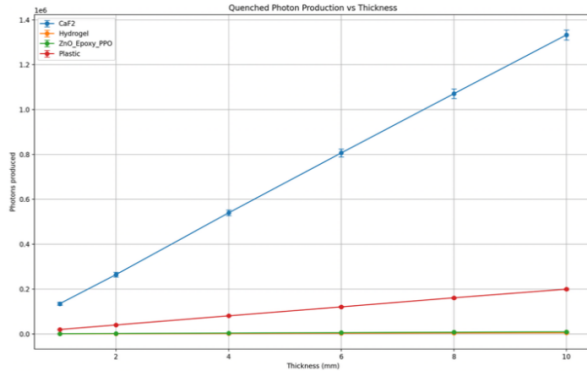


Figure 5.2.3 Quenched Photon Production vs Thickness

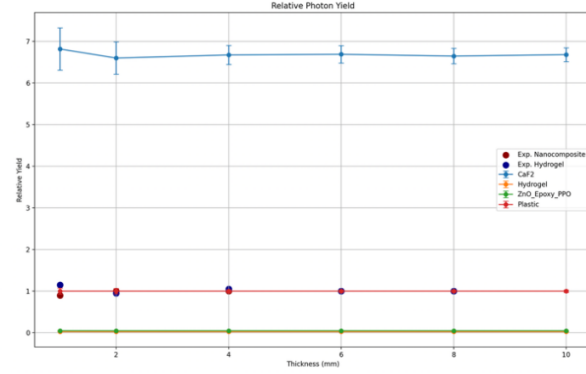


Figure 5.2.4 Relative Photon Yield

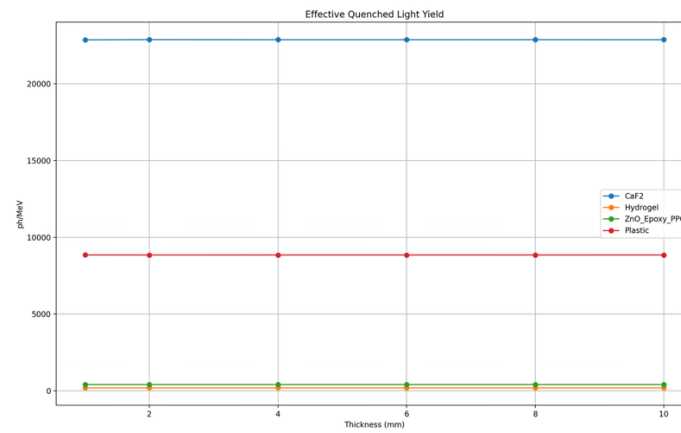


Figure 5.2.5 Effective Quenched Light Yield vs Thickness

The simulation results demonstrate a stable Minimum Ionizing Particle (MIP) plateau. The analysis also shows that the stopping power correlates directly with the material density and the atomic number. $\text{CaF}_2\text{:Eu}$ has the highest stopping power, whereas plastic has the lowest.

Figures 5.2.3, 5.2.4, and 5.2.5 also show that CaF_2 crystals have the highest photon production, relative light yield, and relatively effective quenched light yield. In contrast, hydrogels and nanocomposites demonstrate a lower result for the same parameters.

The observed behaviour aligns with the theoretical expectations. Both deposited energy and photon production increase linearly with thickness, as shown in Figures 5.2.2 & 5.2.3.

5.3 Errors

5.3.1 Experimental Errors

While the experimental setup and simulation attempted to reduce all errors, the method was prone to limitations that can influence the scintillation performance, i.e., the optical effects in the simulation were not accounted for. Despite these limitations, the study produced acceptable results.

6. Discussion and Conclusion

Experimental results indicate that hydrogels and nanocomposites outperform standard plastic scintillators across all parameters. This can be attributed to the material composition lowering the work function wherein the trapped water molecules in hydrogels create a surface dipole layer that further reduces it according to the Helmholtz equation.

$$\Delta\phi = \frac{e\mu n}{\epsilon_0}$$

$\Delta\phi$: Work Function

e : Elementary charge

μ : Dipole moment

n : Surface density

ϵ_0 : Permittivity of free space

However, simulations predict CaF_2 as having the highest photon yield, while hydrogels and nanocomposites have the lowest. This discrepancy highlights the need for high-energy beam testing at CERN to reconcile the theory and experiment.

7. Acknowledgements

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9. Outreach

In the 21st century, interdisciplinary connections are vital to any domain of study, and our research has shown firsthand how essential they are in solving complex problems. While conducting our work, it became evident that breakthroughs often come when physics is applied alongside concepts from chemistry, biology, medicine, and engineering, highlighting the value of thinking beyond traditional boundaries. This realization motivates our outreach program, which aims to inspire students to explore how physics contributes to real-world diverse applications in areas such as renewable energy, medical technologies, and sustainable materials.

Through interactive sessions, hands-on demonstrations, and collaborative problem-solving activities, we encourage students to think creatively, question assumptions, and explore alternative approaches, mirroring the process researchers experience during scientific investigations. Our outreach program aims to help students recognize these interdisciplinary connections and understand the broader impact of physics.

Through our research, we hope to educate and encourage physics enthusiasts to explore alternatives for radiation detection using concepts learned from CERN. By developing hydrogel-based human tissue models and further enhancing their properties, we aim to demonstrate how radiation interacts with biological tissues. These models can serve as valuable and accessible learning tools for both schooling and medical students, while encouraging innovative and sustainable thinking.

